

```
Dataset Shape : (5200, 16)
```

```
Missing Values:
```

```
message_id          0
message_text        0
message_length      0
word_count          0
num_urls            0
num_digits          0
num_special_chars   0
spam_keyword_score   0
legit_keyword_score  0
sender_activity_score 106
sender_account_age_days 113
messages_sent_last_24h 162
timestamp           0
hour_of_day         0
day_of_week         0
spam_label          0
dtype: int64
```

```
===== KNN =====
```

```
K = 3
```

```
Accuracy : 1.0
```

```
K = 5
```

```
Accuracy : 1.0
```

```
K = 7
```

```
Accuracy : 1.0
```

```
===== SVM =====
```

```
===== Naive Bayes =====
```

```
=====
KNN
=====
```

```
Accuracy : 1.0
Precision: 1.0
Recall   : 1.0
F1 Score : 1.0
```

```
Confusion Matrix
```

```
[[845  0]
 [  0 195]]
```

```
Classification Report
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	845
1	1.00	1.00	1.00	195
accuracy			1.00	1040
macro avg	1.00	1.00	1.00	1040
weighted avg	1.00	1.00	1.00	1040

```
=====
SVM
=====
```

```
Accuracy : 1.0
Precision: 1.0
Recall   : 1.0
F1 Score : 1.0
```

```
Confusion Matrix
```

```
[[845  0]
 [  0 195]]
```

```

Classification Report
              precision    recall  f1-score   support

     0           1.00       1.00       1.00        845
     1           1.00       1.00       1.00        195

 accuracy          1.00
 macro avg         1.00
weighted avg         1.00

```

```

=====
Naive Bayes
=====

```

```

Accuracy : 1.0
Precision: 1.0
Recall   : 1.0
F1 Score : 1.0

```

```

Confusion Matrix
[[845  0]
 [ 0 195]]

```

```

Classification Report
              precision    recall  f1-score   support

     0           1.00       1.00       1.00        845
     1           1.00       1.00       1.00        195

 accuracy          1.00
 macro avg         1.00
weighted avg         1.00

```

```

===== Model Comparison =====

```

	Model	Accuracy	Precision	Recall	F1 Score
0	KNN	1.0	1.0	1.0	1.0
1	SVM	1.0	1.0	1.0	1.0
2	Naive Bayes	1.0	1.0	1.0	1.0

```

Best Model:

```

```

Model      KNN
Accuracy   1.0
Precision   1.0
Recall      1.0
F1 Score    1.0

```

```

Name: 0, dtype: object

```

```

PS T:\practice 1234>

```